Riley Brown

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EDUCATION

University of Nebraska-Lincoln
B.S. Mathematics and Data Science, Minor in Computer Science, Lincoln, NE | Aug 2023 - May 2027 (expected)

- GPA 3.92/4.00, University Honors Program
- President of Pi Mu Epsilon and Math Club; Putnam Exam Practice
- D&F Eastman, Regents and Nebraska Career Scholarships; Dean's List every semester

Budapest Semesters in Mathematics
Study abroad, mathematics, Budapest, Hungary | Jan 2026 - Jun 2026

PREPRINTS

Fixed-Parameter Tractability of t-Uniform Hypergraphicality
Riley Brown, Istvan Miklos
arXiv:2606.08523, 2026

RESEARCH EXPERIENCE

Undergraduate Researcher, quantum LDPC codes
University of Nebraska-Lincoln, Lincoln, NE | Fall 2026 | Advisor: Professor Christine Kelley

- Categorizing stopping sets in hypergraph product codes
- Understanding Gallager syndrome decoding outcomes for quantum codes

Research Assistant, pseudorandomness
North Carolina State University, Raleigh, NC | Summer 2026 | Advisor: Professor Chin Ho Lee

- Studied Ta-Shma's epsilon-balanced code construction
- Found simpler constructions and minor improvements to the Hsieh-Mohanty-Zhang construction

Research Assistant, t-uniform hypergraph degree sequences and FPT algorithms
Budapest Semesters in Mathematics, Budapest, Hungary | Spring 2026 | Advisor: Professor Istvan Miklos

- Proved t-uniform hypergraphicality is FPT in (k, t) for compressed degree sequences, extending known bounded-range results
- Reduced realizability to integer programming feasibility via bounded-dimension spectra and balancing hinge-flips

Summer Researcher, algorithm design
Harvey Mudd College, Claremont, CA | Summer 2025 | Advisor: Professor Tim Randolph

- Improved exponential-time algorithms for sumset recognition, an NP-complete problem in additive combinatorics
- Sharpened branching strategies and base-case parameters for graph-theoretic reductions

Undergraduate Researcher, tiling theory
University of Nebraska-Lincoln, Lincoln, NE | Summer 2024 | Advisor: Professor Tri Lai

- Applied Ciucu's Factorization Theorem to decompose tiling enumeration problems into tractable subproblems, working up through independent study of survey papers
- Computed exact tiling counts by applying the Lindstrom-Gessel-Viennot theorem to determinants of binomial coefficient matrices

PROJECTS

leanverify
Lean 4 proof checker against Mathlib: rejects sorries and smuggled axioms | python, lean 4 | 2026

Autoresearch machine
Designed and benchmarked an LLM-driven theorem-proving pipeline (Lean 4 kernel verification, Slurm-scheduled sampling runs) to measure how often models close formal proof statements. | 2026

Deep learning for patient-safety monitoring
Parameter tuning and data analysis for Ocuvera to classify unattended hospital bed exits | python, deep learning | 2025

SKILLS

Programming and tools: Python, MATLAB, Java, SQL, LaTeX, PyTorch, TensorFlow, LangChain, n8n, Lean 4

RELEVANT COURSEWORK

Bijective Partition Theory, **Discrete Mathematics**, **Matroid Theory**, **Advanced Combinatorics**, **Algebra I**, **Coding Theory**, **Development and Analysis of Algorithms**, Graph Theory, Probability Theory, Stochastic Processes, Automata, Computation, and Language, Data Structures and Algorithms